EEEE3116 Coursework 1: Is Lighting Low-Dimensional? — Results Page

Student ID: 20617556 | Code Word: JDNW | Energy Target T = 99.0% | k_T = 5 (Cumulative Energy = 99.65% ≥ 99.0%)

Table 1: Singular Values, Cumulative Energy, and Rank-k Relative Errors (Row k_T = 5 Highlighted)

k	σ_k	Cumulative Energy	Relative Error (Measured)	Relative Error (from σ)
1	121.846	0.918799 (91.88%)	0.284958	0.284958
2	24.3895	0.955612 (95.56%)	0.210684	0.210684
3	18.8741	0.977658 (97.77%)	0.149472	0.149472
4	13.3354	0.988664 (98.87%)	0.106471	0.106471
5 [k_T = 5, ≥ 99.0%]	**11.2873**	**0.996548 (99.65%)**	**0.0587501**	**0.0587501**
6	7.46808	1.000000 (100.0%)	1.40514e-15	0

Table 2: Relative Projection Residuals ||x − p|| / ||x|| of Held-Out Photos H1 (New Lighting) and E1 (Smile)

k	Residual of H1 (New Lighting)	Residual of E1 (Smile)
1	0.36046	0.19214
2	0.355658	0.177788
3	0.3422	0.177693
4	0.327411	0.144526
5	0.322855	0.13118

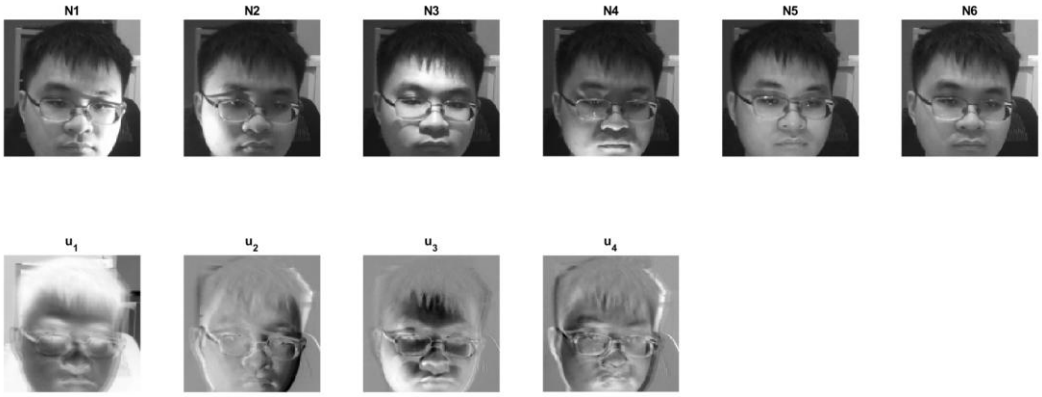


Figure 1: Six neutral lighting photos N1–N6 (top row) and first four left singular vectors u_1–u_4 displayed as images (bottom row).

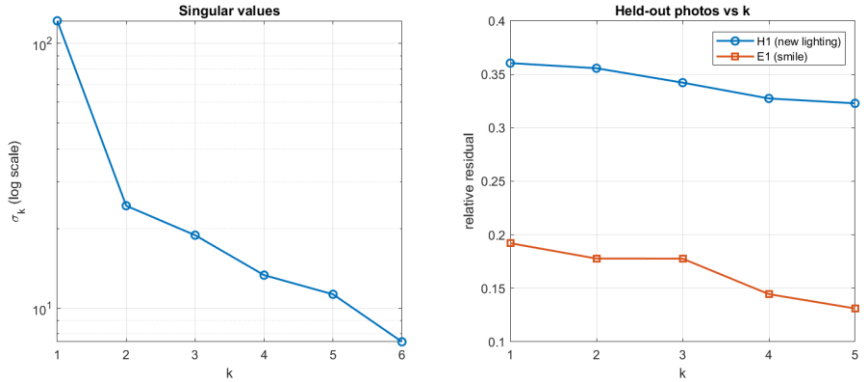


Figure 2: Singular values σ_k on a log scale (left) and relative projection residuals ||x − p|| / ||x|| vs k for H1 and E1 (right).

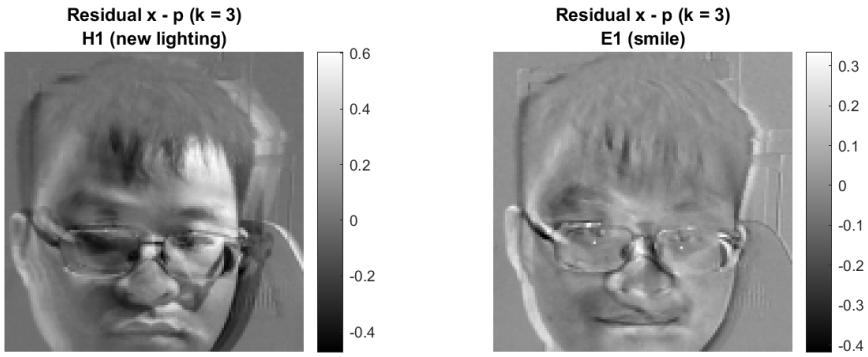


Figure 3: Residual images x − p at k = 3 for held-out photos H1 (new lighting, left) and E1 (smile, right).

AI Declaration: AI tools (Google Gemini) were used to help understand the MATLAB code, review the SVD mathematical derivations, and format the Task 2 results page.